COMP1800

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Introduction

As the famous quote goes, “A picture speaks a thousand words.”, a pictorial representation of any given data can give multiple information in a much more understandable manner without confusion and difficulty. Data visualization is a visualization tool to gain insight from a varied range of data. These tools contain many graphs, charts, plots and infographics to explain complex relationships between data in much clarity, even the hidden details not visible by the naked eye.

In this coursework, the task is to investigate and conclude any characteristics through data visualization with suggestions about the best ideas for ChrisCo, a fictional successful company managing a chain of cinemas across the UK.

Discussions of Visualizations

1. Bar Graph for all Cinemas

Justification

The bar graph was chosen as the first visualization as shown in Figure No.1, to understand the total number of visitors to each cinema in four years. This helps in understanding each cinema's popularity trends or patterns.

Description

The bar graph was plotted with all 30 cinemas to the total number customer visit. The plot was sorted in descending order to illustrate from the most popular cinema to the least popular cinema. The sorted bar graph was then segmented into four parts, namely high, medium, low, and very low popularity. Customer visits above 175000 were high popularity in green colour, between 60000 and 175000 was medium popularity in orange colour, between 25000 and 6000 was low popularity in red colour and below 25000 was very low popularity in black colour. From this, it was concluded that the high popularity cinemas was 'VJV', 'YBS', 'WVA' and 'VPG', medium popular cinemas was 'WBK', 'TPY', 'TJN', 'UVQ', 'UDD', 'SJE', 'RPQ' and 'TVJ', low popular cinemas was 'ACQ', 'WQW', 'XWO', 'XEZ', 'YKT', 'JJQ', 'XQE', 'ZWY', 'WKL' and 'YCI' and the very low popularity cinemas was 'BWF', 'BKK', 'ZQL', 'BQV', 'CCX', 'AKA', 'SDT' and 'CWN'.

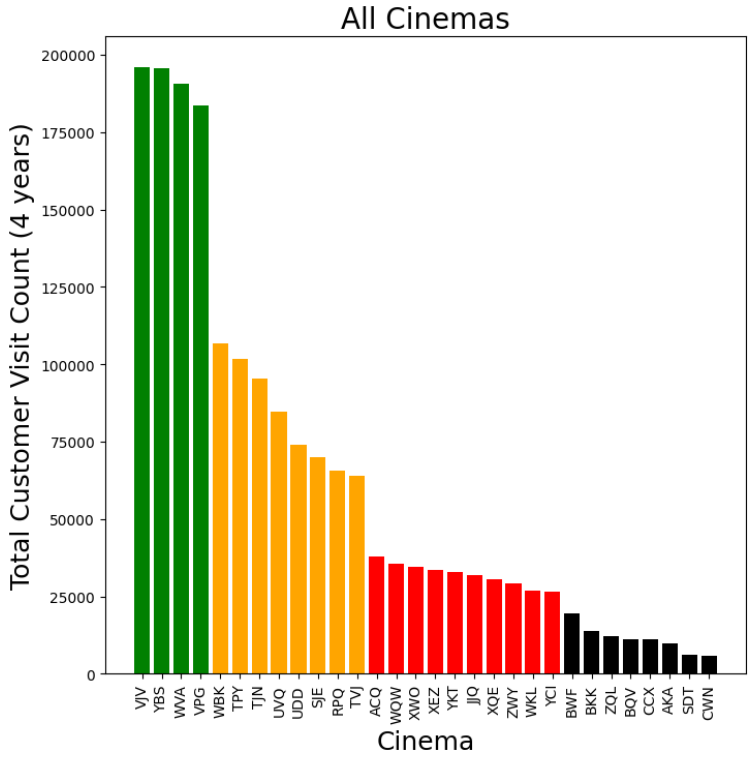

Figure 1. Bar Graph for all cinemas

2. Line Plot for all Cinemas

Justification

The line plot was chosen for all cinemas as the second visualization as in Figure No.2, showing the trend in the frequency of visitors count with respect to the time-period to understand the inclinations and declinations in different time periods.

Description

The line plot was visualized for all the cinemas from January 2019 to January 2023 with respect to the visitors count on each date. The inclusion of trendlines in this plot was to acknowledge the overall trend of each cinema in this interval of time. High popularity cinemas can be recognized from this plot, lying between the visitor count of 750 and 1000. Similarly, medium popularity cinemas can also be recognized which was lying between the visitor count of 250 and 500. Both these cinema categories had a steady horizontal trendline, meaning that the number of visitors visiting these cinemas was stable commonly. Meanwhile, the low and the very low popularity cinema trendlines overlapped over each other and was not difficult to make any inference from this plot.

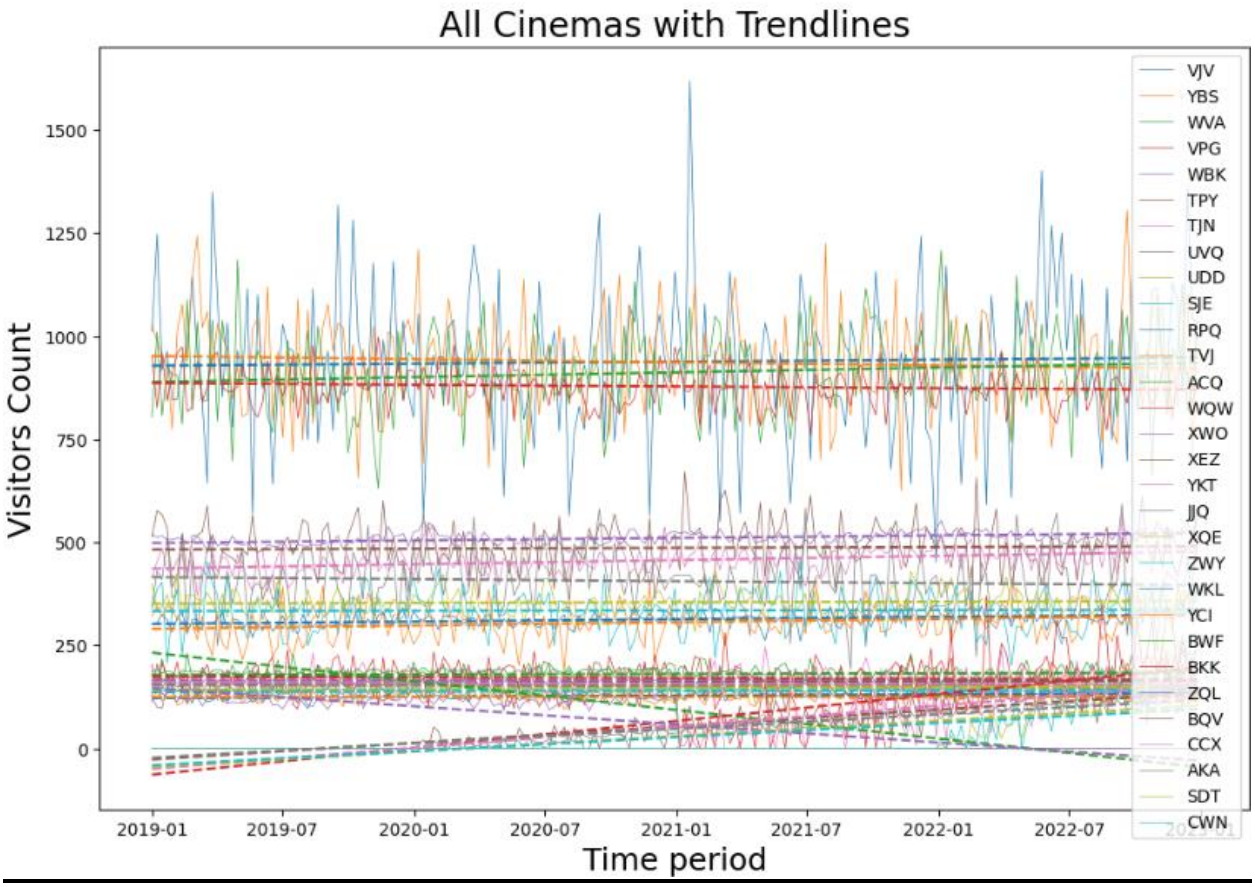

Figure 2. Line Plot for all cinemas

3. Line Plot of Very Low Popularity

Justification

The line plot was chosen for the very low popularity cinemas as the third visualization as in Figure No.3, showing the trend in the frequency of visitors count with respect to the time period to understand the inclinations and declinations in different time periods.

Description

When the line plot for very low popularity cinemas was plotted, all the cinemas in that category were either opening or closing at some point in time. To get a clear pathway of each cinema, a rolling average was included in the plot. Cinemas 'BWF' and 'ZQL' were running constantly with an average visitor count of around 180 and suddenly dropped down to zero visitor count around February 2021, meaning that these two cinemas closed during that period. On the other hand, cinemas 'AKA' and 'BQV', which were closed until January 2020 had opened, growing at a steady pace from there onwards and reached a visitor count of over 100. Similarly, cinemas 'BKK' and 'CCX', which were closed until January 2021 had opened, growing steadily to around 200 and 250 respectively at the end (January 2023). Also, cinemas 'SDT' and 'CWN', which were closed until January 2022 opened with a rapid increase in their visitor count, reaching around 220 and 230 respectively towards the end.

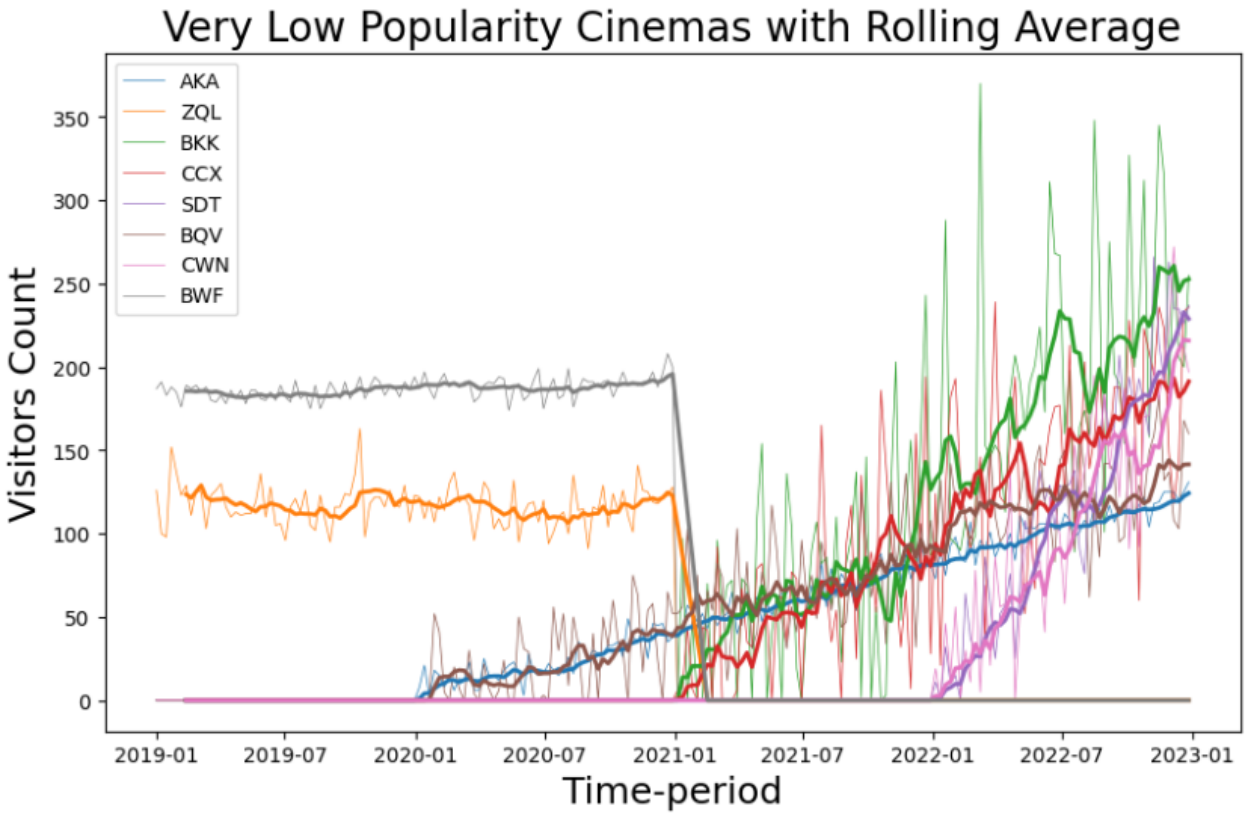

Figure 3. Line plot for Very Low Popularity Cinemas

4. Autocorrelation for all Cinemas

Justification

Seasonality is a pattern that occurs in data in a regular and predictable pattern. The autocorrelation plot was plotted to recognize any seasonality in every cinema.

Description

When every cinema was plotted with an autocorrelation plot, all thirty cinemas as subplots were visualized in a single plot. The seasonality can be proven if the secondary peak crosses the upper dashed lines. From Figure No.4, only the cinema 'WBK' had a regular and predictable pattern, with its secondary peak crossing the upper dashed lines thereby proving its seasonality. The secondary peak is peaking at 50 days making it a quarterly seasonality. All the other plots had irregular and unpredictable patterns. The eight plots at the bottom with thin lines were very low popularity cinemas. All the other plots are plotted inside dashed lines, from which no insights can be proven.

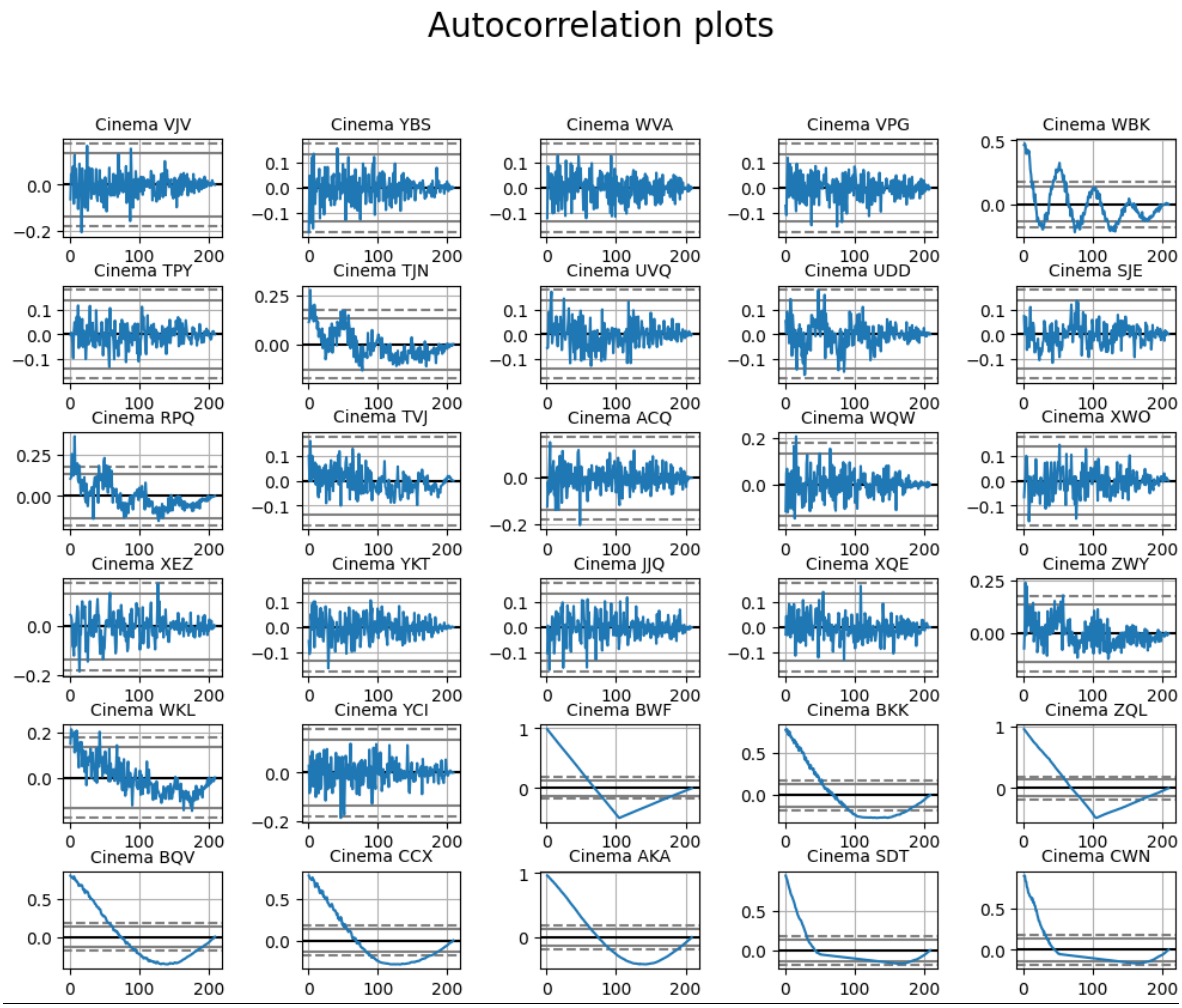

Figure 4. Autocorrelation for all Cinemas

5. Heat Map for Summary Data (Interactive)

Justification

An interactive heat map was chosen for the summary dataset as the fifth visualization as in Figure No.5, showing relationships between each of the summary data such as total visitors, age, capacity, overheads, marketing and spend to find out any co-dependence between them. If any correlation were found, it would be useful for the company to implement better strategies.

Description

The interactive heat map of the summary which includes total visitors, age, capacity, overheads, marketing and spend data was plotted. The positive correlation is denoted by red colour, and the negative correlation is denoted by blue colour. From the map, a very high positive correlation between capacity and total visitors was found. Similarly, a moderate positive correlation between marketing and total visitors, between spend and age and between marketing and capacity was also found. Meanwhile, a slight negative correlation is also found between marketing and spend.

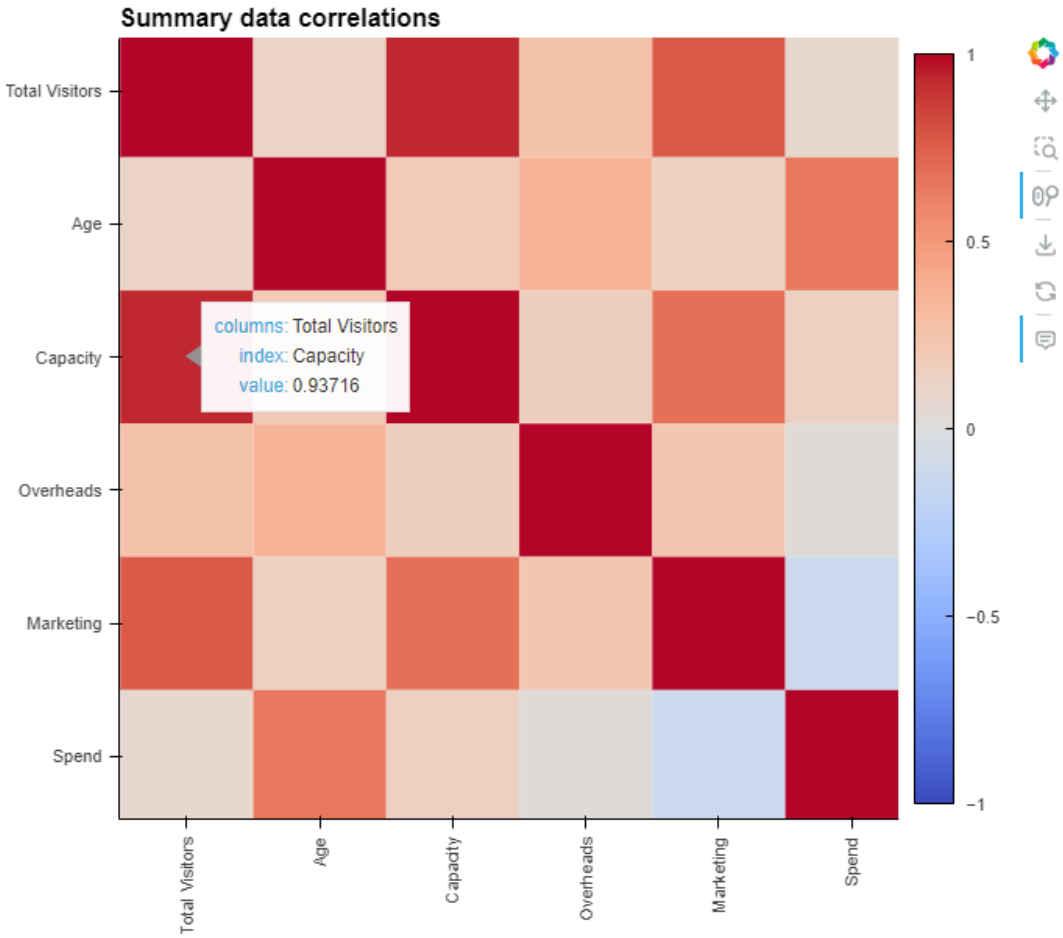

Figure 5. Heat Map for Summary Data (Interactive)

6. Scatter Plot Capacity vs Total Visitors with Linear Regression

Justification

A scatter plot was chosen as the sixth visualization for the relation between capacity and total visitors as shown in Figure No.6, since a very high positive correlation between them was found. A linear regression line was also plotted between the capacity and the total visitors along the best-fit line to check the existence of the outliers, if any.

Description

The scatter plot along with the linear regression was plotted between capacity and total visitors. The capacity and total visitors had a very high positive correlation between them as from the figure. An outlier is an observation which had different characteristics and measures aligning away from the normal pattern. However, there exist no outliers between the seating capacity and the total number of visitors. This means that the number of customers was increasing when the seating capacity of the cinemas was increased perfectly.

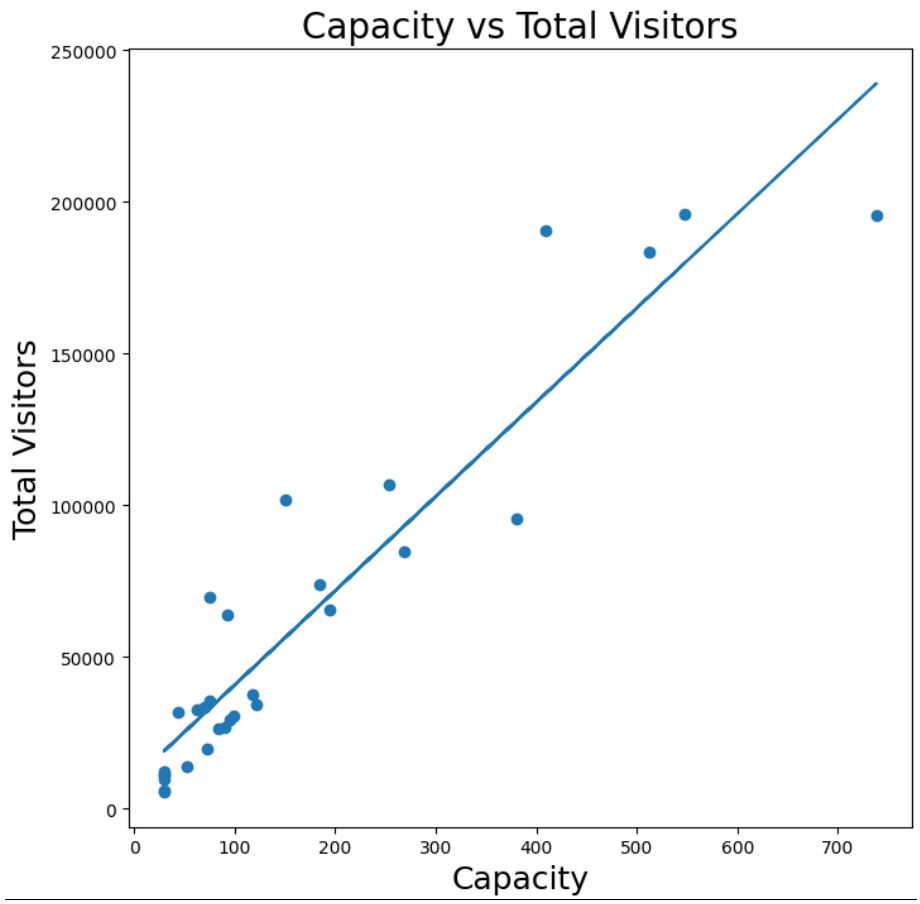

Figure 6. Scatter Plot Capacity vs Total Visitors with Linear Regression

7. Scatter Plot Marketing Spend vs Total Visitors

Justification

A scatter plot was chosen as the seventh visualization for the relation between marketing spend and total visitors as shown in Figure No.7, since a moderate positive correlation between them was found.

Description

The scatter plot was plotted between the marketing spend and the total number of visitors. The marketing spend and the total number of visitors had a moderate positive correlation between them as from the figure. Here, there exist multiple outliers between the marketing spend and the total number of visitors. This means that, even though the number of customers was increasing when the spend amount on marketing the cinemas was increased, multiple outliers were found in the upper central part of the plot. These outliers are the major reason why the correlation between these observations was moderate rather than high correlation.

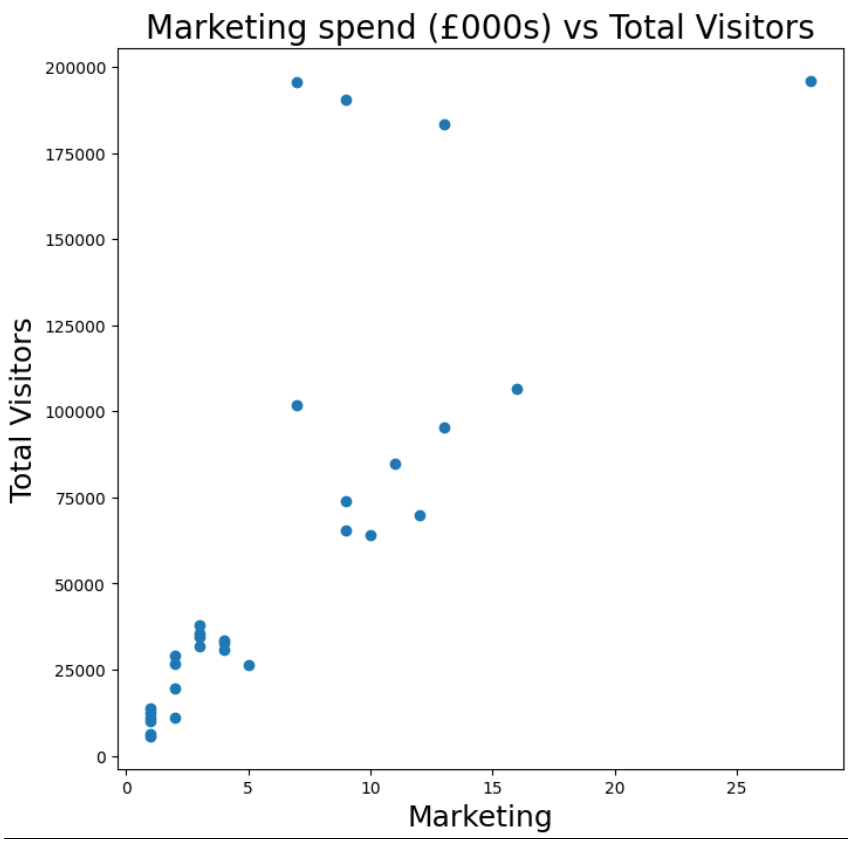

Figure 7. Scatter Plot Marketing Spend vs Total Visitors

8. Box Plot on Summary Data (Interactive)

Justification

The box plot was chosen as the eighth visualization for the summary data as shown in Figure No.8, to understand the skew, symmetry, median, minimum, and maximum value, upper and lower quartiles and outliers in the observation. An interactive plot was plotted to use the zoom tool for the small observations.

Description

The box plot was plotted for the summary data. Since the size of the total visitors count was more than a hundred times bigger than the rest of the observations, it was difficult to get insight from the other observations. So, the visitor count data was removed. The age of viewers had median of 37, minimum and maximum value of 26 and 45 respectively, lower and upper quartile of 30 and 41 respectively, negative skewed and had no outliers. The seating capacity had median of 92, minimum and maximum value of 30 and 380 respectively, lower and upper quartile of 56 and 192 respectively, positive skewed and had four outliers. The average annual overheads had a median of 48, minimum and maximum value of 11 and 99 respectively, lower and upper quartile of 32 and 76 respectively, positive skewed and had no outliers. The average annual marketing spends had median of 4, minimum and maximum value of 1 and 16 respectively, lower and upper quartile of 2 and 9 respectively, positive skewed and had one outlier. The average annual visitor spends had median of 16, minimum and maximum value of 9 and 28 respectively, lower and upper quartile of 11 and 20 respectively, symmetric and had no outliers.

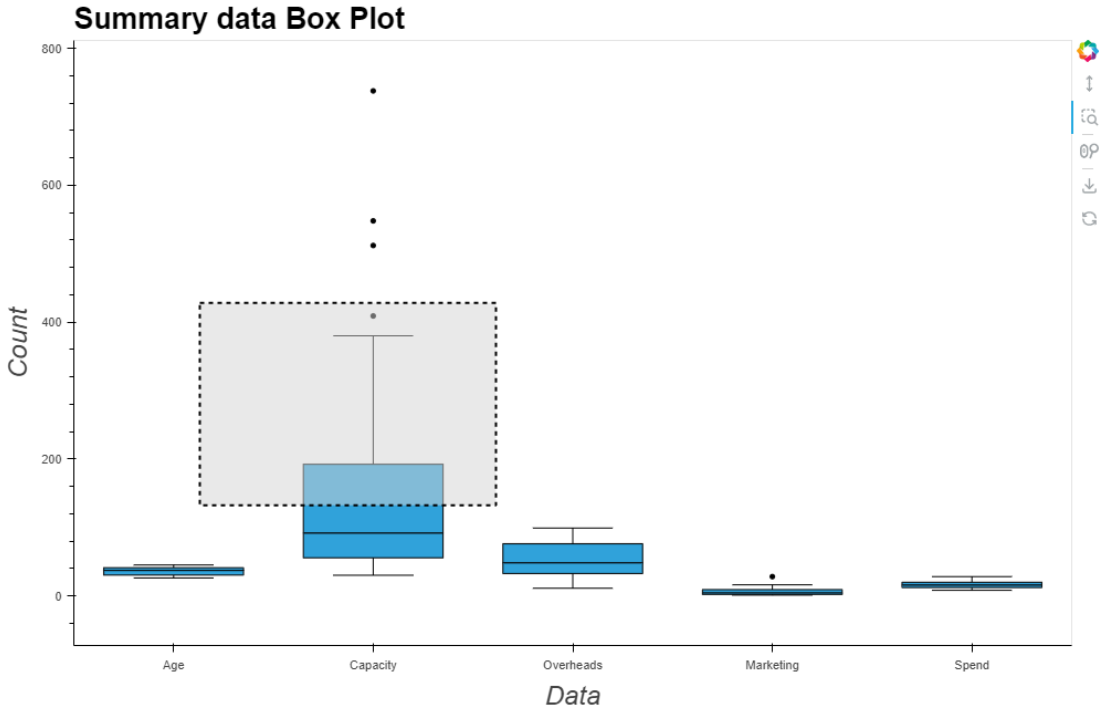

Figure 8. Box Plot on Summary Data (Interactive)

Critical Review

The coursework about ChrisCO's chain of cinema helped me to understand different visualization tools in more depth. Most of the visualization plots such as bar graph, line plots, autocorrelation plot, heat map, scatter plots and box plot were implemented. I also learned the concepts like trendlines, rolling average, seasonality, correlation, linear regression, median, quartiles, skew and outliers throughout the coursework. Trendlines and rolling average gave the tools much more readability, seasonality, correlation and linear regression gave various trends in each observation whereas the rest of them measures of a box plot.

I also learned how to analyze a given data and how to effectively select suitable visualization plots. This helped me to select the best eight visualizations in this coursework. The two interactive visualizations helped me in exploring tools like pan, box zoom, save and reset which helped me find the exact count of each measure in the box plot, for example.

Moreover, writing this report is all about making a non-technical person understand all the explanations without confusion and the need to have technical knowledge. So, I developed a better writing skill when creating this report.

Conclusions

- The High, Medium, Low and very Low popularity categories were plotted to understand the popularity level of each cinema.
- Steady trends in high, medium, and low popularity and varying inclining and declining trends in very low popularity were found.
- The varying trends in very low popularity were checked to find our multiple openings and closings at different time intervals, that is, 2 cinemas closing and 6 cinemas opening.
- The autocorrelation plot showed seasonality for cinema 'BWK' with regular and predictable pattern.
- A very high relation between spend and total visitors and a moderate relation among marketing and total visitors, capacity and marketing and age and spend was observed.
- When the seating capacity in the cinema was increased, the number of viewers visiting the cinema also increased showing.
- When the seating amount spent on marketing was increased, the number of viewers visiting the cinema also increased showing. With multiple outliers existing.
- Capacity and marketing observations had outliers and all the observations were skewed except spend.